Intermediate Elective Option 2

Macey Hartmann

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```r
# general use
library(tidyverse)

# file organization
library(here)

# color display helper
library(prismatic)

# additional scales and formatting
library(scales)

# font support
library(showtext)

# add Bell MT equivalent from Google Fonts
font_add_google("Playfair Display", "playfair")
showtext_auto()
```

```r
# read in vegetation transect data
veg <- read_csv(here("data", "veg.csv"))

# read in pool-level metadata
metadata <- read_csv(here("data", "vp_veg_metadata.csv"))
```

1. Explain your inspiration

I chose Tobias Stalder's "Hiking Locations in Washington" circular bar chart as my inspiration. This visualization uses `coord_polar()` in `ggplot2` to arrange grouped bars radially. His visualization contains bar height, fill color, and an overlaid point and segment.

This format makes sense for the vernal pool vegetation dataset because the data contains multiple discrete pools (analogous to Stalder's hiking regions), each with several continuous summary metrics worth comparing in a single figure. The circular layout compares many pools at once without a cluttered axis and the circular layout feels fitting for a dataset about pools.

My visualization will display the following variables from `veg.csv` and `vp_veg_metadata.csv`:

- **Vernal pool** (`transect_name`): the categorical x-axis, arranged radially
- **Cumulative percent cover** (`sum_pc`): mapped to bar height
- **Number of plant species recorded** (count of distinct psoc values per pool): mapped to fill color via a continuous gradient
- **Mean native percent cover** (filtered from `cover_category == "NATIVE COVER"`): mapped to the point and segment overlaid on each bar

2. Plan your figure

3. Code your figure

```
# create new object from veg
pools_vp <- veg |>
  # filter to only vernal pool transects
  filter(str_detect(transect_name, "VP")) |>
  # complete all combinations of year, pool, species, distance
  # fill missing percent cover values with 0
  complete(year, transect_name, psoc, transect_distance,
           fill = list(percent_cover = 0)) |>
  # group by year, pool, and species
  group_by(year, transect_name, psoc, cover_category) |>
  # calculate total percent cover across all quadrats
  summarize(sum_pc = sum(percent_cover, na.rm = TRUE),
            .groups = "drop") |>
  # create year_pool column from year and transect name, keep originals
  unite("year_pool", year, transect_name, remove = FALSE) |>
  # join metadata to get number of quadrats per pool-year
  left_join(metadata, by = "year_pool") |>
  # calculate mean percent cover from total / number of quadrats
  mutate(mean_pc = sum_pc / num_quad) |>
  # rename transect_name.x back to transect_name after join conflict
```

```r
  rename(transect_name = transect_name.x) |>
  # drop redundant columns brought in by metadata join
  select(-year.x, -year.y, -transect_name.y)

# calculate cumulative cover and mean native cover per pool
pools_vp |>
  group_by(transect_name) |>
  summarise(
    # sum mean percent cover across all years and species per pool
    sum_cover    = sum(mean_pc, na.rm = TRUE),
    # sum of mean percent cover for native species only
    native_cover = sum(mean_pc[cover_category == "NATIVE COVER"],
                       na.rm = TRUE)
  ) |>
  # round native cover to match Stalder's mean_gain rounding
  mutate(native_cover = round(native_cover, digits = 0)) -> summary_stats

# count distinct species with non-zero cover per pool
pools_vp |>
  group_by(transect_name) |>
  summarise(n = n_distinct(psoc[mean_pc > 0])) -> species_counts

# join species counts back to summary stats
left_join(summary_stats, species_counts, by = "transect_name") -> summary_all
```

```r
# color choice
prismatic::color(c("#6C5B7B", "#C06C84", "#F67280", "#F8B195"))
```

```
<colors>
#6C5B7BFF #C06C84FF #F67280FF #F8B195FF
```

```r
# ggplot2
vp_plot <- ggplot(summary_all) +

  # make custom panel grid
  geom_hline(yintercept = 0,   color = "lightgrey") +
  geom_hline(yintercept = 200, color = "lightgrey") +
  geom_hline(yintercept = 400, color = "lightgrey") +
  geom_hline(yintercept = 600, color = "lightgrey") +
  geom_hline(yintercept = 800, color = "lightgrey") +
```

```r
# bars: pool ordered by cumulative cover, filled by species count
geom_col(aes(
  x    = reorder(str_wrap(transect_name, 5), sum_cover),
  y    = sum_cover,
  fill = n),
  position    = "dodge2",
  show.legend = TRUE,
  alpha       = .9) +

# fill gradient and legend title for number of species per pool
scale_fill_gradientn("Number of species",
  colours = c("#6C5B7B", "#C06C84", "#F67280", "#F8B195")) +

# lollipop dot: mean native cover per pool
geom_point(aes(
  x = reorder(str_wrap(transect_name, 5), sum_cover),
  y = native_cover),
  size  = 3,
  color = "gray12") +

# lollipop shaft for native cover per pool
geom_segment(aes(
  x    = reorder(str_wrap(transect_name, 5), sum_cover),
  y    = 0,
  xend = reorder(str_wrap(transect_name, 5), sum_cover),
  yend = 800),
  linetype = "dashed",
  color    = "gray12") +

# annotate bars and lollipops so reader understands scaling
annotate(x     = 4.5,
      y     = 950,
      label = "Total Native\nCover [%]",
      geom  = "text",
      angle = -67.5,
      color = "gray12",
      size  = 5,
      family = "playfair") +
annotate(x     = 9.5,
      y     = 950,
      label = "Cumulative Cover [%]",
      geom  = "text",
      angle = 23,
```

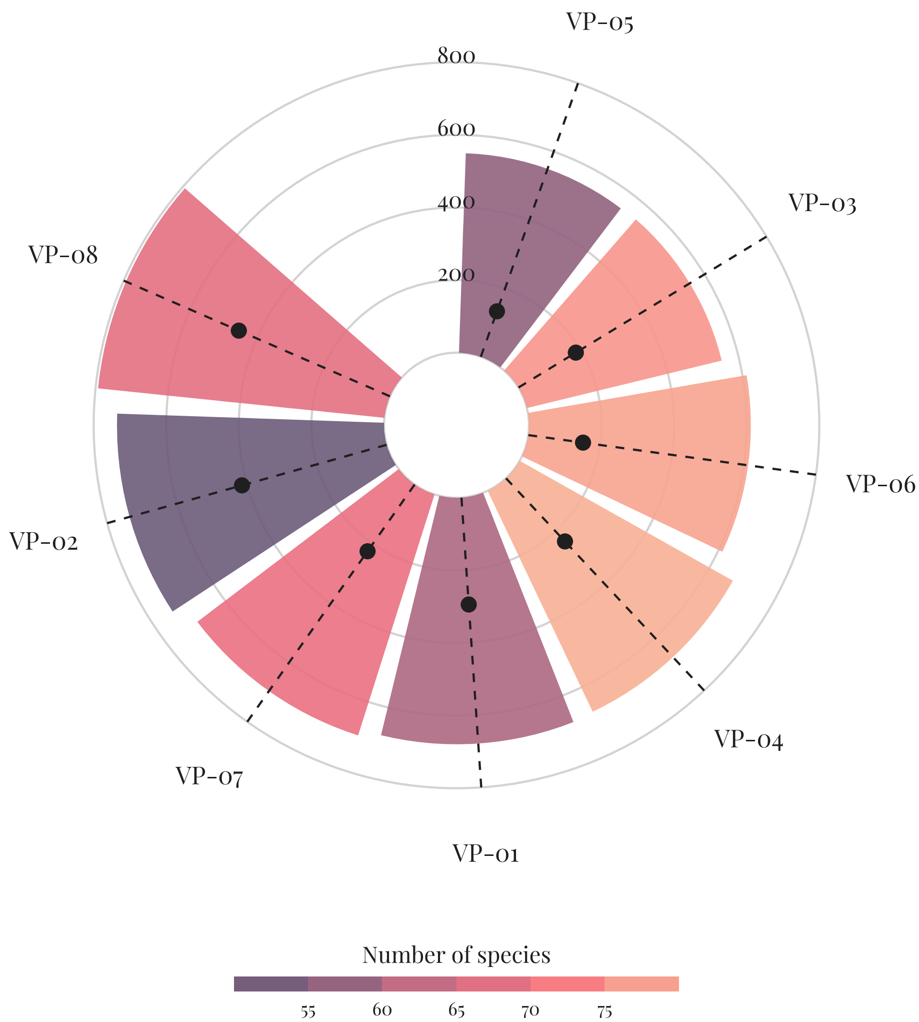
```
       color = "gray12",
       size  = 5,
       family = "playfair") +
annotate(x = 9.7, y = 220, label = "200", geom = "text",
       color = "gray12", family = "playfair") +
annotate(x = 9.7, y = 420, label = "400", geom = "text",
       color = "gray12", family = "playfair") +
annotate(x = 9.7, y = 620, label = "600", geom = "text",
       color = "gray12", family = "playfair") +
annotate(x = 9.7, y = 820, label = "800", geom = "text",
       color = "gray12", family = "playfair") +

# scale y axis so bars don't start in the center
scale_y_continuous(
  limits = c(-200, 850),
  expand = c(0, 0),
  breaks = c(0, 200, 400, 600, 800)) +

# title, subtitle, caption
labs(
  title    = "\nVernal Pool Vegetation",
  subtitle = paste("\nThis visualization shows the cumulative percent
   ↪  cover,",
                   "the number of species and the total native cover per
                    ↪  pool.\n",
                   "VP-08 has the highest cumulative cover,",
                   "while VP-03 has the greatest number of species
                    ↪  recorded.",
                   sep = "\n"),
  caption  = "\nData: Coal Oil Point Reserve vernal pool vegetation
   ↪  monitoring\nVisualization inspired by Tobias Stalder ·
   ↪  tobias-stalder.netlify.app") +

# transform to polar coordinate system
coord_polar() +

# theming
theme(
  legend.position    = "bottom",
  axis.title         = element_blank(),
  axis.ticks         = element_blank(),
  axis.text.y        = element_blank(),
```

```
    axis.text.x        = element_text(color = "gray12", size = 12),
    panel.background   = element_rect(fill = "white", color = "white"),
    panel.grid         = element_blank(),
    panel.grid.major.x = element_blank(),
    text               = element_text(color = "gray12", family = "playfair"),
    plot.title         = element_text(face = "bold", size = 25, hjust =
     ↪  0.05),
    plot.subtitle      = element_text(size = 14, hjust = 0.05),
    plot.caption       = element_text(size = 10, hjust = 0.5)) +

  guides(fill = guide_colorsteps(
    barwidth       = 15,
    barheight      = 0.5,
    title.position = "top",
    title.hjust    = 0.5))

# display figure
vp_plot
```

Vernal Pool Vegetation

This visualization shows the cumulative percent cover,
the number of species and the total native cover per pool.

VP-08 has the highest cumulative cover,
while VP-03 has the greatest number of species recorded.



Data: Coal Oil Point Reserve vernal pool vegetation monitoring
Visualization inspired by Tobias Stalder · tobias-stalder.netlify.app